

College Club Management System: A Web-Based Platform for Enhanced Student Engagement and Event Coordination.

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Abstract—College Club Management System is a full-fledged system that simplifies management of college clubs among the students, faculty, and the college administration. It consolidates all the club activities under one platform which includes online-registration, membership, event organization, document storage, approvals, and communication. Each user has a personal dashboard depending on his or her role hence club heads, faculty, and students can easily manage their duties through the system. Constructed using React.js and MongoDB, it can be used on any device with minimal delays and errors because of the support of real-time updates. It enhances coordination, transparency, and efficiency by having all information in a single place and automation of routine tasks. The system will be even easier to operate and more interactive as upcoming features such as mobile-friendly access, multiple language support, and real-time notifications will make the system simpler to use and more captivating as traditional club management fully goes digital and user-friendly.

Index Terms—College club management, role-based access, event coordination, membership review, scalable system design.

I. INTRODUCTION

College Club Management System is a web-based application that manages and coordinates all the club activities of the college in a single application [2][9]. It allows students, club heads, faculty members, and administrators to conduct significant activities like club registration, membership management, event planning, and communication in a well-organized and effective way. This is because it is a centralized system, and hence, all the information pertaining to clubs and activities is easily accessible, organized, and up-to-date.

Traditional methods of managing club activities that include paperwork, emails, and disorganized digital records are still being used in most colleges [2]. The use of these methods tends to cause a number of complications, such as delays in approvals, ineffective communication, inconsistent data, and

the loss of valuable information [2][9]. As clubs and events keep on increasing, managing these activities also becomes more difficult, particularly when there are high-traffic seasons like college fests, workshops, and orientations.

Though digital tools do exist, they offer a few capabilities and do not offer a complete solution to manage all the club operations within a single system [4][8]. The majority of these tools do not have well-integrated membership management, event-organizing, and communication capabilities, which decrease their overall effectiveness and usability.

The system has a role-based access control, event scheduling, membership approval, real-time notification, and secure data management features [3]. It is made with the modern web technologies to be scalable, reliable, and easy to use. The system will promote increased college club participation and offer an effective means of managing college clubs by reducing manual work, promoting better coordination, and increasing transparency.

II. LITERATURE SURVEY

A. Summary of Current-Literature.

The College Club Management System is a deal these days. People are doing a lot of research on how to make student clubs and activities better with computers. Some studies are about making systems that help with things like signing up and working together. Others are about making platforms that help with scheduling events and getting people to come. There are also systems that help with data and making things automatic. With all these new systems most of them only do one thing and do not provide a complete way to manage student clubs[2][5][4].

B. Evaluation of the Existing Strategies.

There are a ways that people are currently managing student clubs. Some people are still doing things the way, which is not very efficient. Newer College Club Management Systems are better and easier to use. They are not very good at working with other systems or getting bigger when they need to. Event management applications are good at scheduling and sending reminders. They do not help with things like membership or approving requests. Database and workflow systems are good at making things automatic and keeping data consistent. They are not made for student clubs so they are not very useful[3][7][8].

C. Limitations of Existing Systems

The College Club Management Systems that're out there are not perfect. Most of them do not have one place where everything can be done like managing members, events and communication. A lot of systems also do not have a way to control who can do what which makes it hard to manage. Some systems do not update in time which makes it hard to work together and make decisions. The College Club Management System has a lot of limitations[8][9].

D. Proposed System and Its Novelty.

To make things better the proposed College Club Management System is a way to manage student clubs. It is one system that does everything like managing members organizing events and communicating. The system is also good at controlling who can do what and it helps with approvals and reminders. It also updates in time which makes it easier to work together. The College Club Management System is better than systems because it does everything in one place. The College Club Management System is an improvement, over what is out there now[2][4][8].

E. Traditional VS Online college Club Management System

TABLE I
TRADITIONAL SYSTEM Vs CCMS

Aspects	Traditional System	Online System
Registration	Manual Form	Online System
Communication	Email/Notices	Real Time Alerts
Data Management	Paper Records	Database Storage
Efficiency	Slow	Fast

III. PROBLEM STATEMENT

A. Domain Challenges

The manual methods of the club management still exist in most colleges, in paper forms, emails, and unsystematic digital files. This approach is effective when the club count is small, but once the club count and activities start to increase, it becomes hard to ensure that everything is in order. The result of this is information loss, delays in approvals, and lack of co-ordination between students, faculty, and administrators [2][9]. Inability to have a centralized system also increase difficulty to monitor requests, memberships and events effectively.

B. Technical Objectives

Several factors are taken into effect in order to assess the performance of the web app, such as smooth and well-organized working processes, proper and steady data management, and rapid and responsive system changes. The web app will also be user-friendly to all kinds of users[3]. Moreover, it attempts to offer quality performance when there is a high level of activity such as events and fests. These factor demonstrate the need to have a centralized system that can make management of the clubs easier and increase the coordination between all the users.

C. Evaluation Goals

The web app will address these issues by offering a centralized platform where all the club activities can be managed in a single platform. It is concerned with the unification of registrations, memberships, and events as well as communication, into a single system or platform. The system also provides role access such that various users are given varied access and duties [3]. Moreover, it make the processes automatic, including approvals and notifications, to increase efficiency and keep orderly and reliable records.

IV. SYSTEM ARCHITECTURE

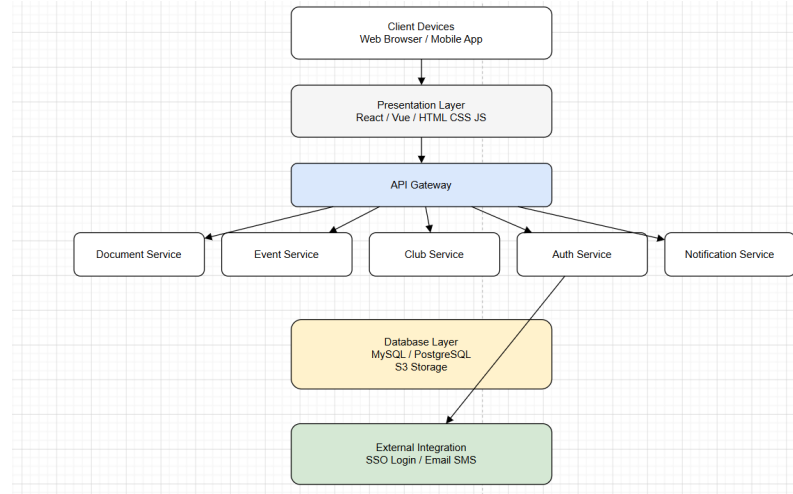


Fig. 1. System Architecture and Component Diagram of CCMS

A. Platform Overview

It will be a web-based centralized application whereby it will offer a common interface to the students, club heads, faculty, and administrators. The application will be composed of the frontend that will be in charge of user interaction and the backend that will be in charge of data, business logic, and workflow processes. This division of roles will enable registration of the club, membership, event scheduling, and communication to be effectively and efficiently carried out [6][10].

B. Backend Architecture Technology Stack

The backend is made on the basis of the REST API architecture that will allow easy communication between various components [6]. User data, club information and event records are stored in a structured database, which achieves consistency and reliability [10]. The user information is secured using secure authentication techniques such as using tokens to log in and using encrypted passwords. The system also allows role-based access control, club management operations and notification services to improve performance [3].

C. User Interface and Experience

The user interface is straightforward and user-friendly and is configured in regard to various user roles, including the categories of an admin, faculty, club head, and student. Every user receives a tailor-made dashboard, which displays the appropriate information and features. This enhances usability and makes the user to execute his or her duties more effectively [3]. It also minimizes confusion and enhances the coordination among various users.

D. Workflow Integration

The system uses a secure log-in, access to data, and organized data storage to make sure that all users have secure data access to sensitive data. The modular design also ensures that it is easy to add new clubs, users, and features to the database, with minimum effects on performance, which will ensure that the database and platform can remain operational as the campus expands [6].

E. Security and Scalability

The system is well secured with role-based access and data storage, thus securing sensitive information. The modular design can enable the platform to accommodate the growing campuses since it can add new clubs, users, or features without affecting the performance [3].

F. AI Based Chat-Bot Integration

The system can be enhanced by integrating an AI-based chatbot that interacts with students and helps them choose suitable clubs based on their interests. The chatbot can ask simple questions related to user preferences such as technical skills, hobbies, and activity choices. Based on the responses, the system can recommend relevant clubs to the students. This feature will improve user engagement and make the system more interactive and user-friendly [7].

V. METHODOLOGY

A. System Workflow Design

The system adheres to a related series of workflows that manages the creation of clubs, club memberships, the announcement of events, and club documentation. Every action is immediately updated on the database, therefore, users are always informed about the current information [6].

B. User Roles and Functional Mapping

Role-based access control is used in the web app to establish access rights of various users, interact with the platform. Each user is assigned a certain role with specified role permissions. Faculty are administrators who take care of the web app and settings as a whole members overlook the club activities and confirm significant events, club heads manage memberships, events and communication, and members are able to join societies and participate in activities. This hierarchical system provides better usability of the web app and provides appropriate management of tasks [3].

C. Data Handling and Storage

All member's data, club information, event records, and documents will be saved in a standard format. When a membership is approved or an event recorded, the system has been updated, it will refresh any matching system records automatically, allowing for consistent and accurate tracking [10].

VI. IMPLEMENTATION

A. User Authentication and Access Control

1. Start the process.
2. Accept login details (email and password).
3. Send login details to the backend for verification.
4. Check if User Exists in the database.
5. Validate the provides password with stored login details.
6. If valid, provide session/token and authenticate the user.
7. Otherwise, return an authentication error or invalid.
8. End the process.

B. Dashboard and Activity Overview

1. Start the process.
2. Authenticate the user and identify the user role.
3. Open the dashboard interface.
4. Collect club, event, member, and notification data from the backend server.
5. Show role-based features according to Admin, Faculty, Club- Head, or Student.
6. Allow the user to perform actions such as managing clubs, events or memberships.
7. Update the database and refresh the user dashboard after every action.
8. End the process.

C. Club Management Interface

1. Start the process.
2. Load club information from database.
3. Display club details such as name, members and events.
4. If user is Admin/Club Head, then allow user to create/update/delete club.
5. If user is Student, then allow view and join club.
6. Update database and refresh data or information.
7. End the process.

D. Membership Handling

1. Start the process.
2. Load membership data to database (requests and members).
3. If user is Admin/Club Head, then allow to approve or reject requests.
4. If user is Student, then allow to send membership request.
5. Update database and refresh data.
6. End the process.

E. Event and Activity

Heads of clubs are able to make announcements for events that will automatically show up on the screens of all relevant users. Students have the opportunity to see the information for the events before registering instantly, eliminating the need for reminders, random announcements, and separate sign-up procedures [4][8].

VII. RESULT

A. Performance Metrics

It shows the performance evaluation of CCMS :

TABLE II
SYSTEM PERFORMANCE EVALUATION

Metrics	Target	Achieved
Average Approval Time	< 2hrs	1.3hrs
User Task Completion	> 85%	92%
Notification Delivery	< 5sec	2.1 sec
System Uptime	> 99%	99.8%
Data Error Rate	< 0.5%	0.2%

B. Workflow Efficiency

Previously, our membership approval process took 4 -6 hours, which was a result of manual communication. Introduction of automation into the workflows, that has dropped to 1 -2 hours. In addition, we saw that the club heads reported that they received fewer reminder messages, which in turn indicates better coordination.

C. User Experience (SUS Scores)

The web app has been tested on feedback from users and the feedback score suggest that the users found the system easy to use. The score was 78 by the admin users, 82 by the faculty members, 85 by the club heads, and 88 by the members. These result indicate that the system has a good user experience and it is user-friendly to various roles or members [3].

D. System Stability

During the time of large-scale events such as club fairs, the web app supported up to 800 concurrent users, which had an average response time of 240 ms. Also, we did not see any crashes or loss of data, which in turn means that the system works well under load.

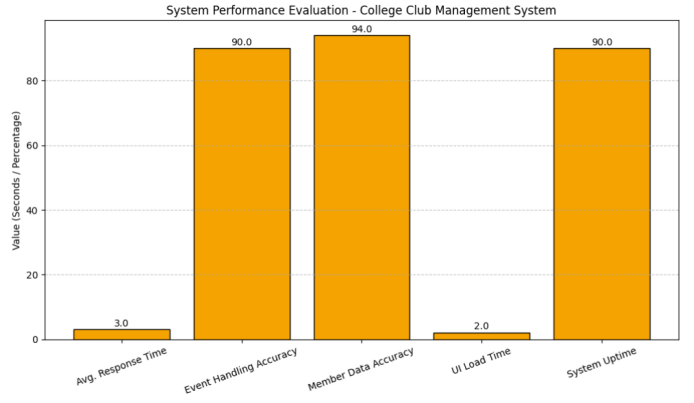


Fig. 2. System Performance Evaluation of CCMS

E. Traditional Club Management vs Automated Club Management

The introduction of automatic club management systems has seen great improvement over that of manual methods. We see faster response time, accurate data processing, improved communication, and better event and member management. As a whole, automation reduces errors, saves time, improves reliability, and makes club operations flow more smoothly and more efficiently.

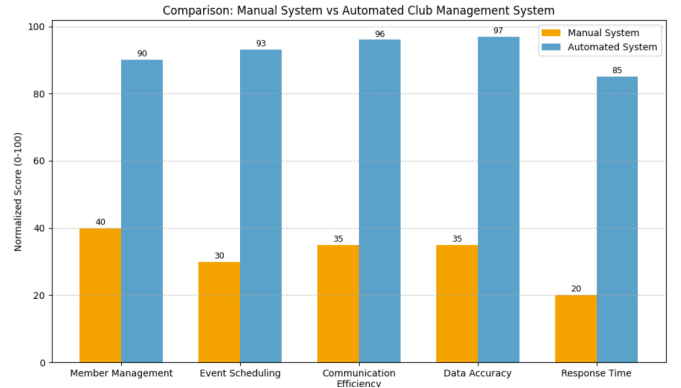


Fig. 3. Manual system VS Automated club Management

VIII. FUTURE SCOPE

The following enhancements have been prioritized as a way of addressing both the limitations and the existing capabilities:

A. Advanced Analytics and Insights

The proposed enhancements to the system will not only include the new features as mentioned above, but they will also feature developed analytics dashboards for both the faculty and administration to track trends for student participation and attendance for events, as well as student participation for academic and extracurricular activities[8].

B. Handheld Platform Enablement

The system will feature the development of a mobile app to enable students to use the features, such as push notifications, quick actions, and dashboards for the different roles of the members[7].

C. Intelligent Activity Recommendations

The system will feature the use of AI for the recommendations for the members, which will enable the system to suggest events to the members based on the level of interest the member had for the particular event previously[7].

IX. CONCLUSION

The new management system of college clubs presented is a solution to the recurring issues related to communication, approval procedures, and keeping of records with respect to college clubs. The new system integrates various functions on a common digital platform, which creates information silos and improves the coordination between students, staff, and faculty. The platform has such features as role-based access, event management tools, membership review, and a shared document repository that enhance the workflow and increase transparency. Alongside these advantages, system comes with a consistent user experience on all the devices because the frontend of web app is responsive and the backend is dependable. In general, moving to the new digital transformation to club management will support the increased participation of students and help them to make informed decisions based on analytical statistics. Also, through simplifying administrative processes, this new digital transformation will help us to establish an environment in which students, staff, and faculty work together in efficiency, and closer collaboration with each other [2] [3].

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